29 September 1966
Superseding
MIL-V-1 8318A (ASG)
8 April 1958

MILITARY SPECIFICATION

VALVE, PRESSURE REGULATING, OXYGEN SYSTEM

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers an oxygen pressure regulating valve.

1.2 Classification - Regulating valves shall be of the following types (see 6.2).

Type I - 70 psi
Type II - 400 psi

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-T-236	Trichloroethylene (Perchloroethylene) Technical Grade
O-T-634	Trichloroethylene, Technical
BB-N-411	Nitrogen, Technical
PPP-T-60	Tape, Pressure-sensitive Adhesive, Vinyl Plastic Film
PPP-B-591	Box, Fiberboard, Woodcleated

FSC 1660

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PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-wall

Military

MIL-P-116	Preservation, Methods of
MIL-C-6541	Chemical Films and Chemical Film Materials For Aluminum and Aluminum Alloys
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-F-22191	Films, Transparent, Flexible, Heat Sealable, For Packaging Applications
MIL-O-27210	Oxygen, Aviators Breathing, Liquid and Gas
MIL-T-27730	Tape, Antiseize, Tetra-fluoroethylene, With Dispenser

STANDARDS

Federal

FED-STD-595	Colors

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Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Specifications And Standards Order Of Precedence For The Selection Of
MS20913	Plug, Square Head Pipe Thread
MS33586	Metals, Definition Of Dissimilar

Air Force-Navy Aeronautical

AND10053	Bosses, Pipe-Thread, Internal-Standard Dimensions For

(Copies of specifications, standards, drawings, and. publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The valve furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 First article - The valve furnished under this specification shall be a product which has been inspected, and passed the first article inspection specified herein.

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3.3. Selection of specification and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials - Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best. quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Metal parts - All metal parts shall be of a corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

3.4.1.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.2 Nonmetallic materials - Any nonmetallic material that is adversely affected by continued use with oxygen shall not be used.

3.4.2.1 Age - Elastomer components, except silicone, shall not be more than 12 months old from the cure date of manufacture to the date of delivery to any Government Service or to any airframe or accessory manufacturer.

3.4.3 Protective treatment - When materials are used in the construction of the valve that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

3,5 Standard parts - MS and AN standard parts shall be used where they suit the purpose, They shall be identified on the manufacturer's drawings by their part numbers, Parts which are not covered by specifications or military standards. or which are not specifically described herein or on the drawings, shall be completely described on manufacturer's drawings,

3.6 Design and construction - The pressure regulating valve shall be designed in accordance with Figure 1. The valve shall be capable of reducing inlet pressures, up to 2,000 psi, to low pressures at the outlet. An adjustable pressure reduction control shall be provided for pre-setting the outlet pressure to a predetermined fixed pressure within the range of 70 to 400 psi.

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3.6.1 Filter - A 200 mesh screen filter or equal shall be firmly secured in the valve inlet. This unit shall be easily removable for cleaning and replacement.

3.6.2 Pressure relief valve - A pressure relief valve, which shall pass the performance requirements specified in 3.7.4., shall be incorporated in the low pressure portion of the pressure regulating valve.

3.6.3 Orientation - The valve shall operate satisfactorily in any position in which the valve is mounted (see 4.6.3).

3.6.4 Antiseize tape - Antiseize tape shall be used on all male pipe thread fittings. Antiseize tape shall conform to and shall be applied as specified in MIL-T-27730. Tape shall not be used on flare tube fitting straight threads, coupling sleeves, or on the outer side of tube flares. None of the tape shall be allowed to enter the inside of a fitting.

3.7 Performance -

3.7.1 Leakage - The valve, when tested as specified in 4.7.2, shall not show any evidence of leakage.

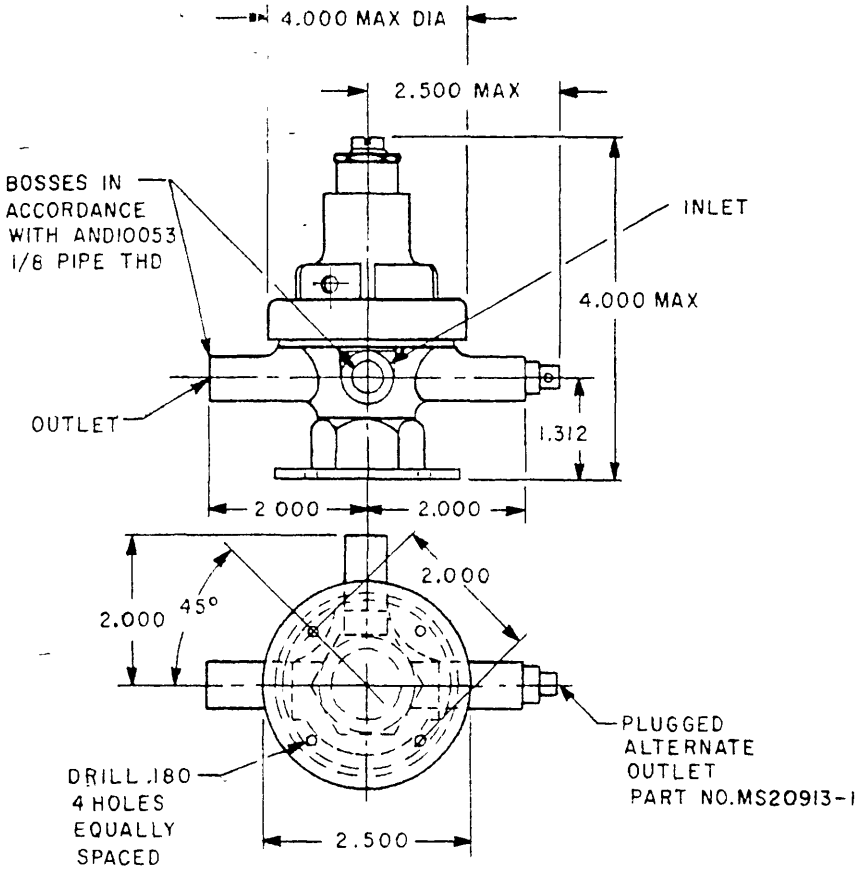
3.7.2 Pressure and flow - The valve, when tested as specified in 4.7.3, shall have the pressure reduction characteristics specified in Table I.

TABLE I

Flow Characteristics

Inlet Pressure (psi)	Outlet pressure with 150 (liters per minute) flow (psi)		Outlet pressure with no flow (psi)	
	Preset at 400 psi	Preset at 70 psi	Preset at 400 psi	Preset at 70 psi
1800	250 (min)	55 (min)	410 (max) 390 (min)	75 (max) 65 (min)
400	250 (min)	55 (min)	410 (max) 390 (min)	75 (max) 65 (min)
300	200 (min)	55 (min)		75 (max) 65 (min)

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DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ±1°, DECIMALS ± .016

FIGURE 1. Valve Dimensions.

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3.7.3 Outlet pressure - The outlet pressure of the valve, when tested as specified in 4.7.4, shall be equal to the preset outlet pressure with the following allowable tolerances:

Outlet pressure range	Tolerance
400 psi	± 10 psi
70 psi	± 5 psi

3.7.4 Pressure relief -

3.7.4.1 Type I - The pressure relief valve, when tested as specified in 4.7.5, shall open when the pressure in the low-pressure side of the pressure regulating valve exceeds 120 psi and the relief valve shall close at not less than 100 psi.

3.7.4.2 Type II - The pressure relief valve, when tested as specified in 4.7.5, shall open when the pressure in the low-pressure side of the pressure regulating valve exceeds 600 psi and the relief valve shall close at not less than 500 psi.

3.7.5 High and low temperature operation - The valve, when tested as specified in 4.7.6, shall operate without failure and shall not show any evidence of mechanical or material failure.

3.7.6 Vibration - The valve, when tested as specified in 4.7.7, shall not show any evidence of mechanical or material failure.

3.7.7 Endurance - The valve, when tested as specified in 4.7.8, shall not show any evidence of mechanical or material failure.

3.7.8 Oxygen bomb test - The pressure regulating valve, when tested as specified in 4,7.9, shall not ignite and there shall not be any evidence of charring or deterioration.

3.8 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.9 Weight - The weight of the pressure regulating valve shall not exceed 1 1/2 pounds.

3.10 Threads -

3.10.1 Screw - Machine screw threads shall be in accordance with MIL-S-7742.

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3.10.2 Pipe - Pipe threads shall be in accordance with MIL-P-7105.

3.11 Color - The color of the valve exterior shall be green, approximately matching color number 14110 of FED-STD-595.

3.12 Finish -

3.12.1 Aluminum and aluminum alloys - Aluminum and aluminum alloys shall be anodized in accordance with MIL-A-8625 or MIL-C-5541.

3.19 Decreasing - All parts of the valve shall be degreased using a vapor phase degreaser in accordance with O-T-236 or O-T-634. Ultrasonics may be used in conjunction with vapor degreasing for the cleaning of components. After completion of the cleaning and when assembled, General Electric Type H Leak Detector or equivalent Halide testing apparatus shall be used to determine the absence of the cleaning compound.

3.14 Identification of product - The valves shall be marked for identification in accordance with MIL-STD-130, except that the Federal Stock Number shall be omitted from the nameplate.

3.14.1 Nameplate - The following information shall be provided on the nameplate:

VALVE , PRESSURE REGULATING , OXYGEN SYSTEM
Specification MIL-V-18318B
Contract or Order No.
Manufacturer’s Part No.
Manufacturer's name or trade-mark

3.15 Workmanship - The valves shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability, or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection - The examination and testing of the valves shall be classified as follows:

(a) Qualification inspection - Qualification inspection consists of examinations and tests performed on samples submitted for approval as qualified products.

(b) First article inspection - First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item meets the requirements of this specification.

(c) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification.

4.3 Qualification inspection - Qualification inspection shall consist of all the examinations and tests of this specification.

4.3.1 Qualification samples - Qualification samples shall consist of two valves. The qualification samples shall be forwarded to the Supply Officer, Naval Air Engineering Center, Philadelphia, Pennsylvania 19112, Attention: Director, Aerospace Crew Equipment Laboratory. The samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-V-18318B and number under authorization (reference authorizing letter and number, see 6.4)

4.4 First article inspection - First article inspection shall consist of the examinations and tests deemed necessary by the testing activity.

4.4.1 First article samples -

4.4.1.1 Applicable to the Naval Air Systems Command - Unless otherwise specified, as soon as practicable after award of a contract or order, the manufacturer shall submit two valves. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a

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contractor is in continuous production of these valves from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the procuring activity. Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article samples shall be forwarded to the Supply Officer, Naval Air Engineering Center, Philadelphia, Pennsylvania 19112, Attention: Director Aerospace Crew Equipment Laboratory, or to a laboratory designated by the procuring activity (see 6.2). The first article samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample submitted by (name) (date) for first article inspection in accordance with the requirements of MIL-V-18318B under contract no.

4.4.1.1.1 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production shall be forwarded to the Government quality control representative. One approved valve shall be returned to the manufacturer for use in monitoring production. The other sample shall be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.4.1.2 Applicable to the Air Force and Army - For purchases by the Air Force and Army, the quantity and kind of samples, the name and address of the first article inspection activity and other pertinent information (see 4.4.1,1 and 4.4.1.1.1 for information) shall be as specified (see 6.2).

4.5 Quality conformance inspection - The quality conformance inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Leakage
Pressure and flow
Outlet pressure
Pressure relief
Preparation for delivery

4.5.1 Sampling -

4.5.1.1 Inspection lot -

4.5.1.1.1 Valve - An inspection. lot size shall be expressed in units of valves, made essentially under the same conditions and from the same materials and components. The sample unit shall be one valve.

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4.5.1.1.2 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing valves fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing valves fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of valves - The sample size, acceptance criteria, tests, and examinations required for the valves shall be as specified in Table II.

4.6 Test conditions -

4.6.1 Temperature and pressure - Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test date, where required, to normal temperature and pressure (NTP) conditions. NTP conditions are 29.92 inches of mercury and 70°F. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4.6.2 Gas - Unless otherwise specified, the gas used in testing the valve shall be oxygen conforming to MIL-O-27210, type I. When specified, nitrogen, conforming to BB-N-411, type I, class I, grade B, or water-pumped air, equivalent in dryness to BB-N-411, type I, class I, grade B nitrogen, may be used provided appropriate density correction factors are applied.

4.6.3 Orientation - When subjected to the tests specified in 4.7.3 and 4.7.4, the valves shall be positioned with the centerline of the passage parallel to a vertical plane. Each consecutive valve tested shall be rotated 90 degrees and 180 degrees about an axis perpendicular to the vertical plane. The rotation shall always be in the same direction.

4.7 Inspection methods -

4.7.1 Visual examination -

4.7.1.1 Valve - Every valve shall be examined visually (for critical defects) to determine conformance to this specification. The classification of defects, Table III, shall be used to classify the defects found.

4.7.1.1.1 Dimensions - The valve shall be checked dimensionally to determine conformance to the dimensions specified herein.

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TABLE II

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATION OF THE VALVES

Inspection	Classification of Characteristic	Method	Sample Size	Acceptance Criteria
Visual Examination (See classification of defects)	Critical Minor	4.7.1.1	Every valve for critical defects. Inspection Level II 1/ for minor defects.	Reject all units with any critical defects. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Dimensions	Major	4.7.1.1.1	Inspection Level S-2 1/	Acceptance number zero, rejection number 1.
Leakage	Critical	4.7.2	Every valve	Reject all defective units
Pressure and Flow	Critical	4.7.3	Every valve	Reject all defective units
Outlet Pressure	Major	4.7.4	Inspection Level S-2 1/	Acceptance number zero, rejection number 1.
Pressure Relief	Critical	4.7.5	Every valve	Reject all defective units
Preparation for Delivery	Minor	4.7.1.2	Inspection Level S-2 1/	Total acceptable quality level of 4.0 percent defective.

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

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TABLE III

CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE VALVES

Critical	Minor
1. Material imperfections-foreign matter embedded	201. Marking-missing, insufficient, incorrect, Illegible, or not permanent.
2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws.	202. Color not as specified.
3. Any component missing, malformed, fractured, or otherwise damaged.	
4. Any component loose or otherwise not securely retained,	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Faulty workmanship or other irregularities.	

4.7.1.2 Preparation for delivery - Each of the fully prepared shipping containers, containing valves selected as a sample unit from the lot shall be examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects, Table IV, shall be used to classify the defects found.

4.7.2 Leakage - The outlet of the valve shall be sealed with a suitable plug. A pressure of 1800 psi shall then be applied to the inlet of the valve and maintained for a period of 15 minutes. The valve shall pass the requirements specified in 3.7.1.

4.7.3 Pressure and flow - The valve shall be preset with an outlet pressure of 400 psi and the inlet shall be connected to a 1800 psi pressure source. The valve shall then be subjected to inlet pressures and flows specified in Table I. The above test procedure shall be repeated with the valve outlet pressure preset at 70 psi. The valve shall pass the requirements specified in 3.7.2.

1/ For qualification inspection, tests shall be conducted as specified in 4.6.3.

2/ For quality conformance inspection, valves shall be tested at an outlet pressure of either 70 psi or 400 psi.

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TABLE IV

LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location. sequence; or method of application; markings not the same on the interior and exterior containers.
Packaging and Packing materials	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements.

4.7.4 Outlet pressure - The valve shall be preset with an outlet pressure in the 400 psi pressure range. A pressure of 1800 psi shall be applied to the valve inlet and a flow of 150 liters per minute shall be maintained through the valve for a period of 30 seconds. The flow shall then be stopped on the outlet side of the valve and the pressure in the outlet side of the valve shall be noted. The above test procedure shall be repeated with the valve outlet pressure preset in the 70 psi pressure range. The valve shall pass the requirements specified in 3.7.3.

1/ For qualification inspection, tests shall be conducted as specified in 4,6.3.

4.7.5 Pressure relief - A pressure of 1800 psi shall be applied to the inlet of the pressure regulating valve. An increasing pressure shall then be applied to the outlet of the valve until the relief valve opens. The relief valve opening pressure shall be noted. The valve outlet pressure shall then be decreased until the relief valve closes. The relief valve closing pressure shall be noted. The valve shall pass the requirements specified in 3.7.4. The valve shall then be subjected to and pass the leakage test.

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4.7.6 High and low temperature operation - A pressure of 1800 psi shall be applied to the inlet of the pressure regulating valve. The valve shall then be subjected to a temperature of 160 ± 5°F for 3 hours. After the conditioning period and while still at this temperature, the valve shall be subjected to and pass the outlet pressure test. The valve shall then pass the requirements specified in 3.7.5. The valve shall then be returned to room temperature and shall be subjected to and pass the leakage test and the pressure and flow test. The above test procedure shall be repeated utilizing a temperature of -65 ± 5°F for 3 hours. After the conditioning period and while still at this temperature, the valve shall be subjected to and pass the outlet pressure test. The valve shall then pass the requirements specified in 3.7.5. The valve shall then be returned to room temperature and subjected to and pass the leakage test and the pressure and flow test.

4.7.7 Vibration - A pressure of 1800 psi shall be applied to the inlet of the pressure regulating valve. The valve shall then be vibrated for six hours at a frequency from 300 to 3000 cycles per minute and a double amplitude of 0.018 to 0.020 inches. The valve shall be cycled six times, each cycle being one hour, and consisting of varying the frequency from 300 to 3000 and back to 300 cycles per minute. The valve shall be mounted in each of the three mutually perpendicular axes of vibration for a two hour period in each plane. During the vibration period, a flow of 30 ± 5 liters per minute shall be maintained through the valve, The valve shall pass the requirements specified in 3.7.6. The valve shall then be subjected to and pass the leakage test and the pressure and flow test.

4.7.8 Endurance - The valve shall be connected to a 1800 psi oxygen pressure source. The inlet pressure to the valve shall be varied from 1800 psi to 300 psi by means of a suitable cycling device or control valve. This shall constitute one cycle. The length of the cycling period shall be a minimum of 60 seconds. During the cycling period, an outlet flow of 30 liters per minute shall be maintained through the valve. The valve shall be subjected to a minimum of 1000 cycles. Upon completion of this test procedure, the valve shall be subjected to and pass the leakage and the pressure and flow tests.

4.7.9 Oxygen bomb test - The valve assembly shall be subjected to oxygen bomb tests to determine compatibility of the item with high pressure oxygen.

4.7.9.1 Equipment - The equipment used in performing these tests shall be in accordance with Figure 2. The main valve shall be a rapid-opening 1/4 inch valve, shall require no internal lubrication for operation, and shall be rated for at least 3,000 psi operation. The valve shall be leak tight when closed and shall open with such rapidity that at least 95 percent of the total pressure rise shall have occurred within the test chamber in 20 ± 7 milliseconds. The pressure rise time may be determined

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by measurement with a fast-responding small volume pressure transducer substituted for the valve in the test setup. The tubing and fitting connecting the valve to the test chamber shall be 3/16 inch size and shall be fabricated from corrosion-resistant steel. The run of tubing between the valve and the test chamber shall not be bent more than 4 degrees. The bleed valve shall be of a type which uses no organic material to accomplish its seal and requires no interior lubrication for operation. The bleed valve shall be installed so that when it is closed, the test chamber does not communicate with the packing side of the valve.

4.7.9.2 Procedure (valve assembly) - The outlet connections of the valve shall be sealed. The valve inlet shall be installed in the test setup shown on Figure 2. The quick-acting valve shall be fully opened and the valve pressurized with oxygen to a pressure of 2000 psig. This pressure shall be maintained for 30 seconds and then released through the bleed valve. The valve shall be disconnected and examined for indication of charring or deterioration of any nonmetallic material. The valve shall pass the requirements specified in 3.7.8.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A - Unless otherwise specified, each valve shall be preserved in accordance with MIL-P-116, Method 1C-4. No contact preservative is required. Each valve shall be sealed within a snug fitting bag of 0.0015 inch gage material conforming to MIL-F-22191, Type III. The sealed valve shall be cushioned and packaged within a waterproof fiber can with sealed metal ends. The valve shall be cushioned in such a manner as to prevent damage to itself. Six unit packages shall be packaged within a fiberboard container conforming to PPP-B-636, Type I, Weather Resistant Class, Grade V3C or V3S. The manufacturer's seam shall be adhesive sealed. All seams and joints shall be sealed with tape conforming to PPP-T-60.

5.1.2 Level C - The valve shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing - Packing shall be Level A, B, or C, as specified (see 6.2).

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5.2.1 Level A - Unless otherwise specified, sixty valves, packed as specified in 5.1,1, shall be packed in overseas type exterior containers conforming to PPP-B-591, PPP-B-636, or PPP-B-640, weather resistant Class. Closure and strapping of the containers shall be in accordance with the applicable container specification or appendix thereto. The gross weight of the packed shipping container shall not exceed the weight limitations of the applicable container specification.

5.2.2 Level B - Sixty valves, packaged as specified in 5.1.1, shall be packed in domestic type exterior containers conforming to PPP-B-591, PPP-B-636, or PPP-B-640, Domestic Class. Closure and strapping of the containers shall be in accordance with the applicable container specification or appendix thereto. The gross weight of the packed shipping container shall not exceed the weight of the packed shipping container shall not exceed the weight limitations of the applicable container specification.

5.2.3 Level C - The packaged valves, which the carrier requires to be packed, shall be packed within exterior type shipping containers in a manner that shall insure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking - Nomenclature tags shall be inserted in each bag showing title, part number or other data as required by procuring activity. In addition, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking - The following precautionary marking shall appear on each package:

" CAUTION: DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT THE OXYGEN VALVES"

6. NOTES

6.1 Intended use - The oxygen pressure regulating valve covered by this specification is intended for use in aircraft oxygen systems for reducing the high pressure (1800 psi) from oxygen cylinders to the lower working pressure of oxygen-dispensing equipment.

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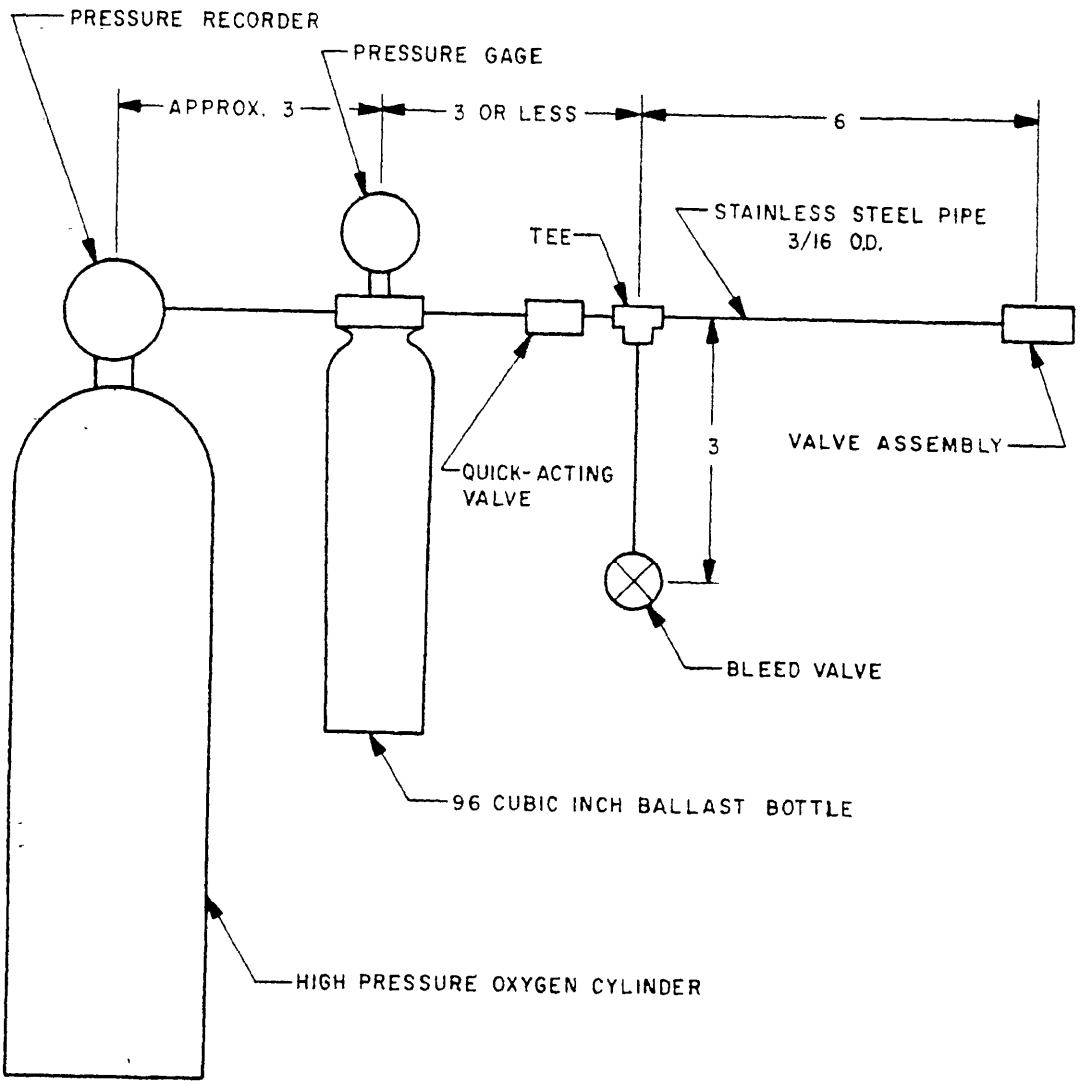


FIGURE 2 Oxygen bomb equipment

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6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Whether first article inspection is required and where the samples should be delivered (see 4.4.1.1 and 4.4.1.2).
(d) Selection of applicable levels of preservation, packaging, and packing.
(e) Applicable methods of cleaning and preservation.
(f) Items of data required (see 6.3).

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NavWeps Form 4200/15(Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.4 With respect to products requiring qualification, awards will be made only for such products as have prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Commander, Naval Air Systems Command, Washington, D.C., 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Engineering Center, Philadelphia, Pennsylvania, 19112, Attention: Director, Aerospace Crew Equipment Laboratory. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Aerospace Crew Equipment Laboratory indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

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Custodians:

Army - M O
Navy - AS
Air Force -71

Preparing Activity:

Navy - AS

Review Activities:

Army - MO
Navy - AS
Air Force -11 and 71

Project No. (1660-0045)

NOTICE: Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-V-18316B Valve, Pressure Regulating, Oxygen System

ORGANIZATION *(of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Weapons Engineering Standardization Office (Code
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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